

Answer any 01 out of 03 Questions

1. Find out Cosine similarity among the documents of a self prepared Term-Frequency Vector (Four documents will be in the document vector).
2. How we can compute the dissimilarity between two binary attributes? Explain with suitable example.
3. What is five-number summary of a distribution? Explain the relevant example using Boxplots.

Patuakhali Science and Technology University

7th Semester (Level-4, Semester-I), Mid Examination of B.Sc. Engg. (CSE), January-June: 2021

Course Code: CCE-417 Course Title: Data Warehousing and Mining

Credit Hour: 3.0 Full Marks: 15

1. How can we compute the dissimilarity between two binary attributes? 5
2. Define the following terms with example: 5
 - i. Nominal Attributes
 - ii. Ratio-scaled attribute
3. Consider the following database with five transactions. Let $\text{min sup count} = 2$ and $\text{min conf} = 50\%$. 5

Transaction	Item Occurrence
T1	A, B, C
T2	A, C, D
T3	A, B, C, D
T4	A, D, E
T5	B, C

- (a) Find all frequent itemsets using Apriori.
- (b) List all the *strong* association rules.

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Final Examination of B. Sc. Engineering in CSE Level: 4 Semester: I Session: 2017-2018

Course Code
CCE-417

Course Title
Data Warehousing and Mining

January-June 2021

Credit: 03
Time: 03 Hr
Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1. [A.] Define the following terms with example. 7
i. Discrete attributes ii. Continuous attributes
iii. Nominal Attributes iv. Ratio-scaled attribute

- [B.] How we can compute the dissimilarity between two binary attributes? Explain with suitable example. 7

2. [A.] a. Define the following terms with appropriate example: 7
i. min-max normalization
ii. Stepwise backward elimination
b. Find out similarity of diseases among Jack, Jim and Mary using the following patient record table (Table 1) which contains the attributes *name*, *gender*, *fever*, *cough*, *test-1*, *test-2*, *test-3*, and *test-4*, where *name* is an object identifier.

Table 1
Relational Table Where Patients Are Described by Binary Attributes

name	gender	fever	cough	test-1	test-2	test-3	test-4
Jack	M	Y	N	P	N	N	N
Jim	M	Y	Y	N	N	N	N
Mary	F	Y	N	P	N	P	N
:	:	:	:	:	:	:	:

- [B.] Find out Cosine similarity among the documents of a self-prepared Term-Frequency Vector (Four documents will be in the document vector). 7

3. [A.] a. What is five-number summary of a distribution? Explain with a suitable example using Boxplots. 7
b. "Values have a meaningful order but magnitude between successive values is not known"—justify the above statement.

- [B.] a. How do you measure proximity for ordinal attributes? Describe with a suitable example. 7
b. Compare between data matrix and dissimilarity matrix.

4. [A.] Why is *naïve Bayesian classification* called "naïve"? Briefly outline the major ideas of naïve Bayesian classification. 5

- [B.] Both *k-means* and *k-medoids* algorithms can perform effective clustering. 4
a. Illustrate the strength and weakness of *k-means* in comparison with *k-medoids*.
b. Illustrate the strength and weakness of these schemes in comparison with a hierarchical clustering scheme (e.g., *AGNES*).

- [C.] How the ensemble method called "Random Forests" works? 3

- [D.] What is outlier? Define various types of outliers with example. 2

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- 5 [A.] Suppose that the data mining task is to cluster points (with (x, y) representing location), where the points are: P1(7,4), P2(6,4), P3(5,6), P4(4,2), P5(6,3), P6(5,2), P7(3,3), P8(4,5), P9(6,5), P10(3,6), P11(4,4), P12(8,2). The distance function is Euclidean distance. Part of the distance matrix is given below (Table 2).

Table 2

P1	0												
P2	1	0.00											
P3	2.8	2.24	0.00										
P4	2.6	2.83	4.12	0.00									
P5	2.9	1.00	3.16	2.24	0.00								
P6	2.8	2.24	4.00	1.00	1.41	0.00							
P7	4.1	3.16	3.61	1.41	3.00	2.24	0.00						
P8	2.1	2.24	1.41	3.00	2.83	3.16	2.24	0.00					
P9	1.4	1.00	1.41	3.61	2.00	3.16	3.61	2.00	0.00				
P10	4.0	3.61	2.00	4.12	4.24	4.47	3.00	1.41	3.16	0.00			
P11	2	2.00	2.24	2.00	2.24	2.24	1.41	1.00	2.24	2.24	0.00		
P12	2.2	2.83	5.00	4.00	2.24	3.00	5.10	5.00	3.61	6.40	4.47	0.00	
	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	

- Plot the points. Find the hidden distances.
 - With $Eps=1.9$ and $MinPts=3$, find clusters using DBSCAN.
 - Show the core, noise and border points.
- [B.] Suppose that you have generated two classification models, M1 and M2, from any given data. You have performed 10-fold cross-validation to obtain a mean error rate for each. How can you determine which model is best? What if any difference between the two may just be attributed to chance? 4
- [C.] How to Generate Association Rules from Frequent Itemsets? 3
- [D.] Make comparison between closed and maximal frequent itemsets. Give examples. 2
- 6 [A.] Consider the following dataset (Table 3). Find the first stump using AdaBoost algorithm. Use Gini Index for internal calculation. 5

Table 3

Row No.	Gender	Age	Income	Illness
1	Male	41	40000	Yes
2	Male	54	30000	No
3	Female	42	25000	No
4	Female	40	60000	Yes
5	Male	40	50000	Yes

- [B.] Consider table 4 which shows the probability value (column 3) returned by a probabilistic classifier for each of the 10 tuples in a test set, sorted by decreasing probability order. Column 1 is merely a tuple identification number, which aids in explanation. Column 2 is the actual class label of the tuple. Now, plot the ROC curve for the given data. 4

Table 4

Tuple #	Class	Prob.
1	P	0.90
2	P	0.80
3	N	0.70
4	P	0.60
5	P	0.55
6	N	0.54
7	N	0.53
8	N	0.51
9	P	0.50
10	N	0.40

- Make comparison between CF tree and dendrogram with example.
- [C.] BIRCH applies a *multiphase* clustering technique. What are the primary phases? 3
- Or
- What is the main strategy behind CLIQUE for identifying a candidate search space?
- [D.] Define the general approaches for *improving* the classification accuracy of class-imbalanced data. 2

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CCE-417

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Data Warehousing and Mining

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Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1. [A.] Find out dissimilarity of diseases among Jack, Jim and Mary using the following patient record table (Table 1) which contains the attributes *name*, *gender*, *fever*, *cough*, *test-1*, *test-2*, *test-3*, and *test-4*, where *name* is an object identifier.

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Mary	F	Y	N	P	N	P	N
:	:	:	:	:	:	:	:

- [B.] Find out Cosine similarity among the documents of a self-prepared Term-Frequency Vector (Four documents will be in the document vector).

2. [A.] "Values have a meaningful order but magnitude between successive values is not known"—justify the above statement.

[B.] Illustrate the various methods we can use to manage data.

3. [A.] i. Let $x_1 = (1, 2)$ and $x_2 = (3, 5)$ represent two objects. Find the Euclidean, Manhattan and Supremum distance.

ii. Compare between a data matrix and a different matrix.

[B.] How is dissimilarity computed between objects described by nominal attributes?

4. [A.] How we can compute the dissimilarity between two binary attributes? Explain with suitable example.

[B.] i. What is the difference between a symmetric and an asymmetric binary?
ii. Compare between discrete and continuous attributes.

5. [A.] Illustrate the following terms with example:

- Stepwise forward selection
- Stepwise backward elimination
- Combination of forward selection and backward elimination

[B.] How do you measure the proximity of ordinal attributes? Describe with a suitable example.

6. [A.] What is five-number summary of a distribution? Explain the relevant example using Boxplots.

[B.] Define the following terms with appropriate example:

- min-max normalization
- z-score normalization
- normalization by decimal scaling

3. (A) Define the following terms with example

- Nominal Attributes
- Symmetric binary attribute
- Interval-scaled attributes

4(B) How we can compute the dissimilarity between two binary attributes? Explain with suitable example.

5(A) i. Let $x_1 = (1, 2)$ and $x_2 = (3, 5)$ represent two objects. Find out Euclidean, Manhattan and Supremum distance

ii. Define the following terms with appropriate example:

- min-max normalization
- z-score normalization
- normalization by decimal scaling

5(B) How we can measure dissimilarity among the documents of the following table.

Document Vector or Term-Frequency Vector

Document	team	coach	hockey	baseball	soccer	penalty	score	wln	loss	season
Document1	5	0	3	0	2	0	0	2	0	0
Document2	3	0	2	0	1	1	0	1	0	1
Document3	0	7	0	2	1	0	0	3	0	0
Document4	0	1	0	0	1	2	2	0	3	0

6(A) Suppose that a patient record table (Table 1) contains the attributes *name*, *gender*, *fever*, *cough*, *test-1*, *test-2*, *test-3*, and *test-4*, where *name* is an object identifier. Find out similarity among Jack, Jim and Mary.

Table 1
Relational Table Where Patients Are Described by Binary Attributes

name	gender	fever	cough	test-1	test-2	test-3	test-4
Jack	M	Y	N	P	N	N	N
Jim	M	Y	Y	Z	Z	Z	Z
Mary	F	Y	N	...	...	...	...
...	...	...	...	...	...	...	...

6(B) Explain the different approaches which we can use for handling missing data.

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